$$\begin{cases} \rho u_{tt} - (\mu u_x)_x = f & \text{in } (0,L) \times (0,T] \\ u_t(x,0) = v_0(x) & x \in (0,L) \\ u(x,0) = u_0(x) & x \in (0,L) \\ u(0,t) = g_1(t) & t \in (0,T] \\ u(L,t) = g_2(t) & t \in (0,T] \end{cases}$$

$\rightarrow u_1$

- $u_1 = \rho u_t \qquad u_2 = \mu u_x$

1) $(u_1)_t - (u_2)_x = f$

- $(u_1)_x = \rho u_{tx} \qquad (u_2)_t = \mu u_{xt} = \mu u_{tx}$

2) $(u_2)_t = \frac{\mu}{\rho} (u_1)_x$

EQ: $$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}_t + \begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}_x = \begin{pmatrix} f \\ 0 \end{pmatrix}$$

$$\underline{U} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \qquad A = \begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix}$$

$= \pm c_1$

$\text{eig}(A) \Leftrightarrow |A - \lambda I| = 0 \qquad \lambda^2 - \frac{\mu}{\rho} = 0 \qquad \lambda_{1,2} = \pm\sqrt{\frac{\mu}{\rho}}$

IC. $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}(x,0) = \begin{pmatrix} \rho v_0(x) \\ \mu u_0(x) \end{pmatrix}$

$$u_t(x,0) = \frac{u_1(x,0)}{\rho} = v_0(x)$$

$$u_2(x,0) = \mu u_x(x,0) = \mu (u_0)_x$$

B.C. $u(0,t) = g_1(t)$

$$u_t(0,t) = g_1'(t) \iff \overbrace{\rho u_t(0,t)}^{u_1(0,t)} = \rho g_1'(t)$$

$$\begin{cases} u_1(0,t) = \rho g_1'(t) \\ u_1(L,t) = \rho g_2'(t) \end{cases}$$

$$A = T^{-1} D T \qquad D = \begin{pmatrix} \sqrt{\frac{\mu}{\rho}} & 0 \\ 0 & -\sqrt{\frac{\mu}{\rho}} \end{pmatrix}$$

$$T = \left( \underline{w}_1 \mid \underline{w}_2 \right)$$

$$A \underline{w}_1 = \sqrt{\frac{\mu}{\rho}}\, \underline{w}_1$$

$$\begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix} \begin{pmatrix} w_1^1 \\ w_1^2 \end{pmatrix} - \sqrt{\frac{\mu}{\rho}} \begin{pmatrix} w_1^1 \\ w_1^2 \end{pmatrix} = 0$$

$$\begin{cases} -w_1^2 - \sqrt{\mu/\rho}\, w_1^1 = 0 \\ -\frac{\mu}{\rho} w_1^1 - \sqrt{\frac{\mu}{\rho}}\, w_1^2 = 0 \end{cases} \qquad w_1^2 = -\sqrt{\frac{\mu}{\rho}}\, w_1^1$$

$$-\frac{\mu}{\rho} w_1^1 + \frac{\mu}{\rho} w_1^1 = 0 \qquad 0 = 0$$

$$\begin{cases} w_1^1 = t \\ w_1^2 = -\sqrt{\frac{\mu}{\rho}}\, t \end{cases} \qquad t = 1 \Rightarrow \begin{cases} w_1^1 = 1 \\ w_1^2 = -c \end{cases}$$

$$\underline{w}_1 = \begin{pmatrix} 1 \\ -c \end{pmatrix} \frac{1}{\sqrt{1+c^2}} \qquad \lambda = c_1$$

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$$\begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix} \begin{pmatrix} w_2^1 \\ w_2^2 \end{pmatrix} + \sqrt{\frac{\mu}{\rho}} \begin{pmatrix} w_2^1 \\ w_2^2 \end{pmatrix} = 0$$

$$\begin{cases} -w_2^2 + \sqrt{\frac{\mu}{\rho}}\, w_2^1 = 0 \\ -\frac{\mu}{\rho} w_2^1 + \sqrt{\frac{\mu}{\rho}}\, w_2^2 = 0 \end{cases} \qquad w_2^2 = +\sqrt{\frac{\mu}{\rho}}\, w_2^1 \qquad -\frac{\mu}{\rho} w_2^1 + \frac{\mu}{\rho} w_2^1 = 0$$

$$\underline{w}_2 = \begin{pmatrix} w_2^1 \\ w_2^2 \end{pmatrix} = \begin{pmatrix} 1 \\ c \end{pmatrix} \frac{1}{\sqrt{1+c^2}} \qquad \lambda = -c$$

$$\|\underline{w}_2\| = 1$$

$$\boxed{\begin{aligned} T &= \left( \underline{w}_1 \,\middle|\, \underline{w}_2 \right) = \begin{pmatrix} 1 & 1 \\ -c & c \end{pmatrix} \frac{1}{\sqrt{1+c^2}} \\ T^{-1} &= \sqrt{1+c^2} \cdot \frac{1}{2c} \begin{pmatrix} c & -1 \\ c & 1 \end{pmatrix} \end{aligned}}$$

$$TT^{-1} = I = \frac{1}{2c} \begin{pmatrix} 1 & 1 \\ -c & c \end{pmatrix} \begin{pmatrix} c & -1 \\ c & 1 \end{pmatrix}$$

$$= \frac{1}{2c} \begin{pmatrix} 2c & 0 \\ 0 & 2c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\underline{W} = T^{-1} \underline{U} = \frac{\sqrt{1+c^2}}{2c} \begin{pmatrix} c & -1 \\ c & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$= \frac{\sqrt{1+c^2}}{2c} \begin{pmatrix} cu_1 - u_2 \\ cu_1 + u_2 \end{pmatrix}$$

$$\underline{W}_t + D\,\underline{W}_x = T^{-1}\begin{pmatrix} f \\ 0 \end{pmatrix}$$ ← DECOUPLED SYSTEM

$$W_1 = \frac{\sqrt{1+c^2}}{2c}\left(c u_1 - u_2\right)$$

$$W_2 = \frac{\sqrt{1+c^2}}{2c}\left(c u_1 + u_2\right)$$

$$\begin{cases} (W_1)_t + \sqrt{\frac{\mu}{\rho}}\,(W_1)_x = \frac{\sqrt{1+c^2}}{2\cancel{c}}\,\cancel{c}\,f \\ (W_2)_t - \sqrt{\frac{\mu}{\rho}}\,(W_2)_x = -\frac{\sqrt{1+c^2}}{2}\,f \end{cases}$$

$$W_1 + W_2 = \frac{\sqrt{1+c^2}}{\cancel{c}}\,\cancel{c}\,u_1 = \underline{\sqrt{1+c^2}}\,u_1$$

$$\sqrt{1+c^2}\,u_1(0,t) = \overset{\sqrt{1+c^2}}{\sqrt{\rho}}\,\rho_1'(t) = W_1(0,t) + W_2(0,t)$$

$$\sqrt{1+c^2}\,u_1(L,t) = \overset{\sqrt{1+c^2}}{\sqrt{\rho}}\,\rho_2'(t) = W_1(L,t) + W_2(L,t)$$

$$\underbrace{\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}}_{\text{COUPLE } W_1 \,\&\, W_2 \text{ ON THE BOUNDARY}} \begin{pmatrix} W_1(0,t) \\ W_2(0,t) \\ W_1(L,t) \\ W_2(L,t) \end{pmatrix} = \begin{pmatrix} \rho\, g_1'(t) \\ \rho\, \rho_2'(t) \end{pmatrix} \sqrt{1+c^2}$$

# HOW TO SOLVE THIS SYSTEM

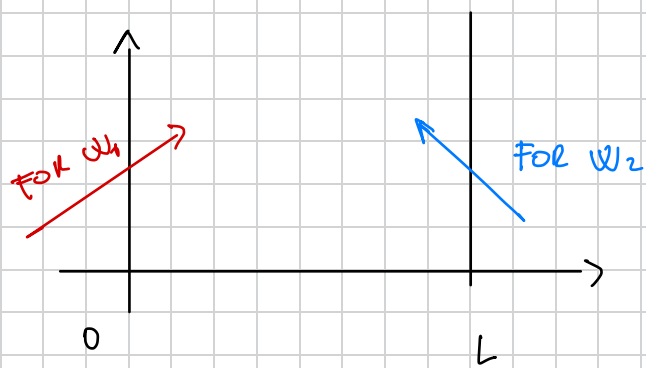
1) Compute $T$ & $T^{-1}$ $\qquad \underline{w} = T^{-1}\underline{U}$

2) $\underline{w}_t + D\,\underline{w}_x = \underline{G}$ $\qquad \left( \underline{G} = T^{-1}\begin{pmatrix} f \\ 0 \end{pmatrix} \right)$

WITH MY FAVOURITE SCHEME (FD or FEM)

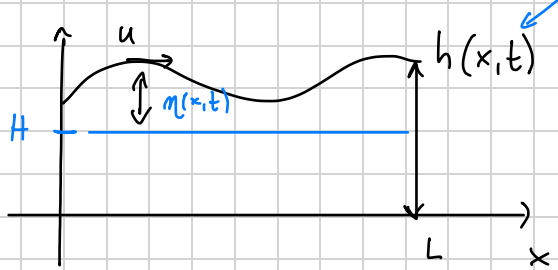
3) POSITIVE EIG $\quad w_1(0, t^{n+1}) = \rho\sqrt{1+c^2}\, g_1'(t) - w_2(0, t^{n+1})$

NEG. EIG $\quad w_2(L, t^{n+1}) = \rho\sqrt{1+c^2}\, g_2'(t) - w_1(L, t^{n+1})$




4) $\underline{U} = T\,\underline{w} = \begin{pmatrix} \rho u_t \\ \mu u_x \end{pmatrix}$

# SHALLOW WATER EQs




WATER HEIGHT

$$h(x,t) = H + \eta(x,t)$$

FREE SURFACE DISPLACEMENT

MEAN DEPTH

$u(x,t)$ HORIZONTAL VELOCITY OF PARTICLES

$$\begin{cases} h_t + (hu)_x = 0 \\ (hu)_t + \left(hu^2 + \frac{1}{2} g h^2\right)_x = 0 \end{cases}$$

NON LINEAR SYSTEM

GRAVITY CONSTANT

LINEARIZE AROUND THE STEADY STATE

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = 0 \end{cases}$$

$q = Hu$

$$\underline{U} = \begin{pmatrix} \eta \\ q \end{pmatrix}$$

$$\begin{pmatrix} \eta \\ q \end{pmatrix}_t + \underbrace{\begin{pmatrix} 0 & 1 \\ gH & 0 \end{pmatrix}}_{A} \begin{pmatrix} \eta \\ q \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\mathrm{eig}(A) = \pm \underbrace{\sqrt{gH}}_{c}$$

c WAVE VELOCITY

HOW TO SOLVE HYPERBOLIC SYSTEM WITH A "STAGGERED APPROACH"

- $v = u_1 = \rho u_t \qquad \sigma = u_2 = \mu u_x$

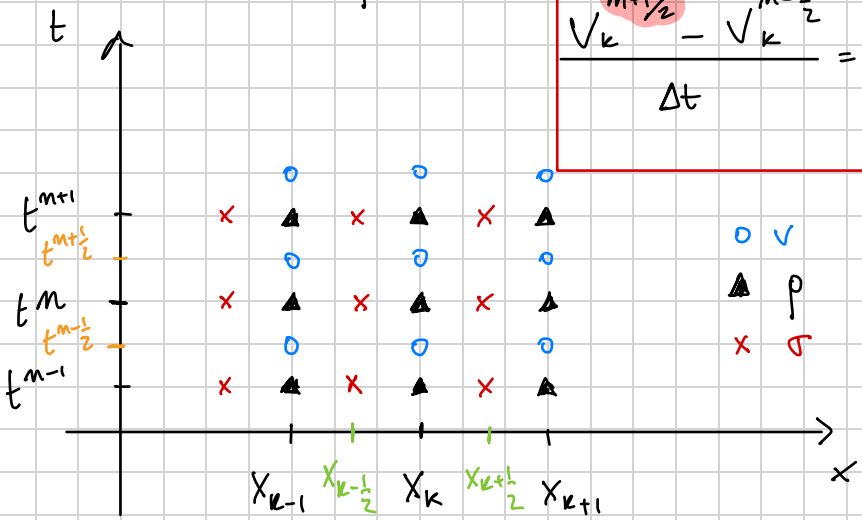
$$\begin{pmatrix} v \\ \sigma \end{pmatrix}_t + \begin{pmatrix} 0 & -1/\rho \\ -\mu & 0 \end{pmatrix} \begin{pmatrix} v \\ \sigma \end{pmatrix}_x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} v_t - \frac{1}{\rho}\sigma_x = 0 & \text{1 EQ} \\ \sigma_t - \mu v_x = 0 & \text{2 EQ} \end{cases}$$

$$\rightarrow \rho v_t - \sigma_x = 0$$

1: EQ $\quad v_t - \frac{1}{\rho}\sigma_x = 0$

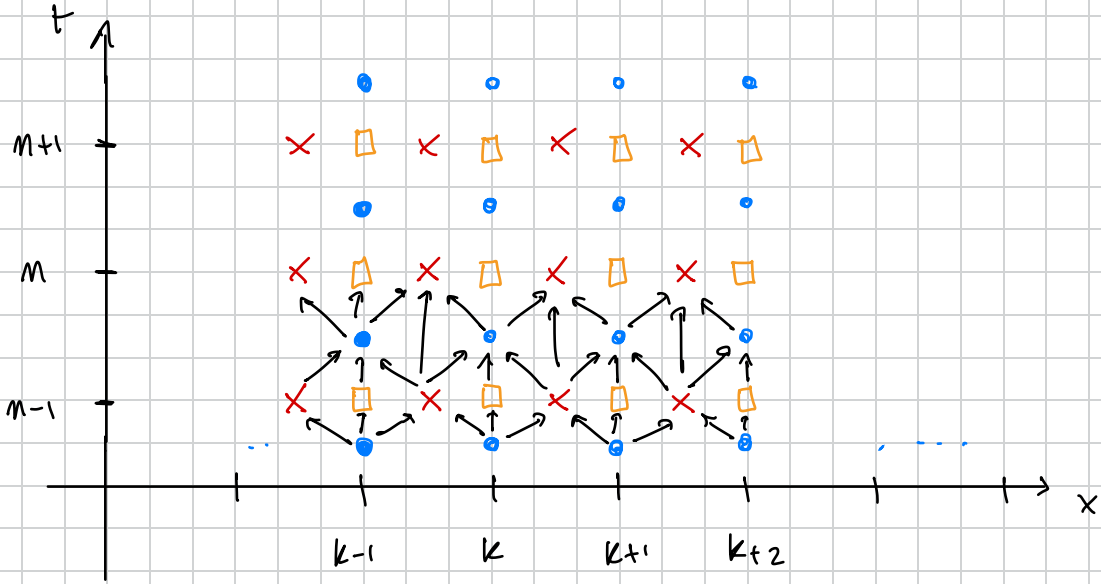
$$\boxed{\frac{V_k^{n+1/2} - V_k^{n-\frac{1}{2}}}{\Delta t} = \frac{1}{\rho_k}\,\frac{\sigma^n_{k+\frac{1}{2}} - \sigma^n_{k-\frac{1}{2}}}{\Delta x}}$$




2EQ $\quad \sigma_t - \mu v_x = 0$

$$\boxed{\frac{\sigma^{n+1}_{k+\frac{1}{2}} - \sigma^n_{k+\frac{1}{2}}}{\Delta t} = \mu_k\,\frac{v^{n+\frac{1}{2}}_{k+1} - v^{n+\frac{1}{2}}_k}{\Delta x}}$$

$$\begin{cases} v_k^{n+\frac{1}{2}} = V_k^{n-\frac{1}{2}} + \dfrac{1}{\rho_k}\,\dfrac{\Delta t}{\Delta x}\left(\sigma^n_{k+\frac{1}{2}} - \sigma^n_{k-\frac{1}{2}}\right) & \forall k \\ \sigma^{n+1}_{k+\frac{1}{2}} = \sigma^n_{k+\frac{1}{2}} + \mu_k\,\dfrac{\Delta t}{\Delta x}\left(v^{n+\frac{1}{2}}_{k+1} - v^{n+\frac{1}{2}}_k\right) & \forall k \end{cases}$$




STAGGERED

$$\sigma \rightsquigarrow \left( \frac{\Delta x}{2}, \Delta t \right) \qquad + \text{B.C.}$$

STAGGERED

$$v \longrightarrow \left( \Delta x, \frac{\Delta t}{2} \right) \qquad + \text{I.C.}$$

I.C.

$$\begin{cases} v(x,0) = v_0(x) \\ \sigma(x,0) = \sigma_0(x) \end{cases}$$

ATTENTION

$$v_k^{n+\frac{1}{2}} = \dots \qquad v_k^0 = ???$$

$$\sigma_{k+\frac{1}{2}}^{n+1} = \sim \sigma_{k+1/2}^0$$

GOOD

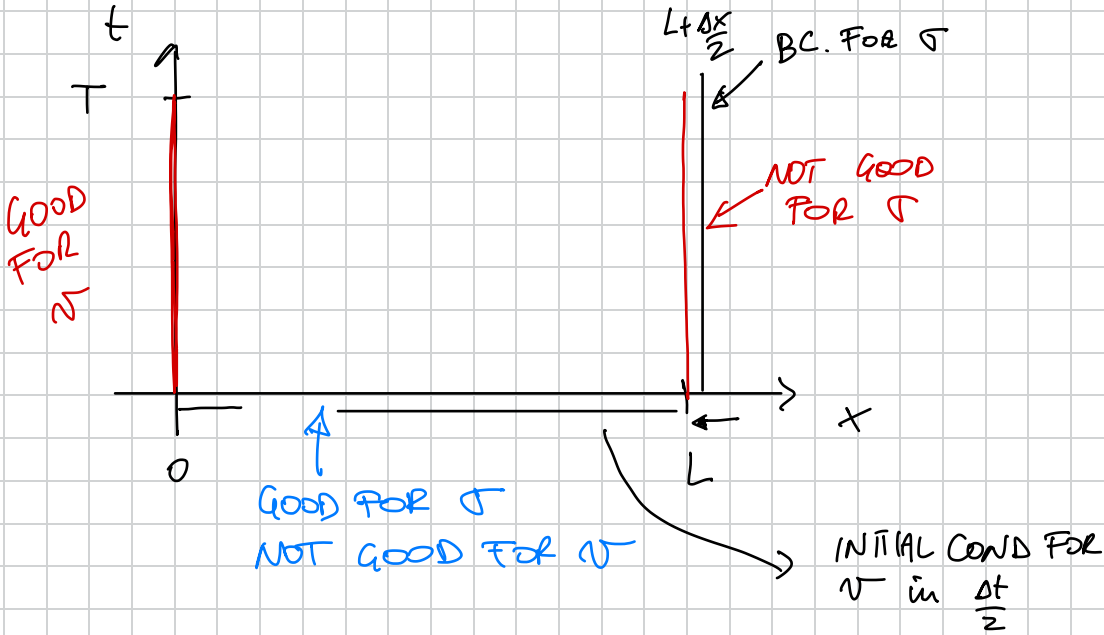
B.C.

$$\begin{cases} v(0,t) = \psi(t) \\ \sigma(L,t) = \varphi(t) \end{cases}$$

$v_0^{n+\frac{1}{2}}$ GOOD

$\sigma_{k+\frac{1}{2}}^{n+1}$ NOT GOOD

IS NOT ALIGNED WITH BOUNDARIES




I.C. FOR $v$ are given in $t^{-\frac{1}{2}}$

$v(0,-\Delta t) = \dots$

$\sigma(L+\frac{\Delta x}{2}, t) = \varphi(t)$

# SCALAR NON LINEAR HYPERBOLIC EQs.

$$u_t + F(u)_x = 0$$

$F(\cdot)$ IS CALLED FLUX

$$\underline{U}_t + [\underline{F}(\underline{U})]_x = 0$$

$F(u) = a u$ → TRANSPOR EQ.

$\underline{F}(\underline{U}) = A \underline{U}$ → HYPERBOLIC SYSTEM OF EQs (CONSERVATION LAWS)

If $x \in (a, b)$

$$\int_a^b u_t \, dx + F(u(b)) - F(u(a)) = 0$$

$$\frac{d}{dt} \underbrace{\int_a^b u \, dx}_{\text{CONSTANT OVER TIME}} = 0$$

$F(u) = a u$ TRANSPORT EQ

$F(u) = \frac{1}{2} u^2$ BURGER'S EQ. (SIMPLIFICATION OF N-S eq.)

CONSERVATIVE FORM

$$u_t + F(u)_x = 0$$

$$\downarrow$$

$$u_t + \left[\frac{1}{2}u^2\right]_x = 0$$

NON-CONSERVATIVE FORM

$$u_t + F'(u)\, u_x = 0$$

$$\downarrow$$

$$u_t + u\, u_x = 0$$